

**DATA ANALYTICS WITH POWER BI
PROJECT REPORT**

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CRIME LENS DASHBOARD

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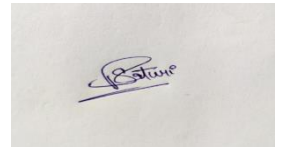
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DECLARATION

I, Samridhi Raturi, student of BTech under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.



Date: 14-12-2025

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LINKS

LinkedIn:

https://www.linkedin.com/posts/samridhiraturi17_crimelens-powerbi-dataanalytics-ugcPost-7405932086878044160-wUW8?utm_source=share&utm_medium=member_desktop&rcm=ACoAAGJlfAYBV4ZmmiqUfk-geqHBI22EJo3I4ak

Google Drive:

<https://drive.google.com/drive/folders/1ySuGv6GjuJZOhi5YkDjvEL8MxHfjuqMn?usp=sharing>

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DATA SCIENCE MINOR PROJECT REPORT

TITLE: Crime Lens (Interactive Crime Analysis Dashboard using Power BI)

1. Introduction

Crime has a direct impact on the safety, well-being, and development of society. With the rapid growth of urbanization and population, the volume of crime-related data has increased significantly. Analyzing this large amount of data manually is time-consuming and inefficient, which makes data-driven tools essential for modern crime analysis. Data science and business intelligence techniques play a crucial role in transforming raw crime data into meaningful insights that can support law enforcement agencies and policymakers.

The objective of this project, **Crime Lens – Interactive Crime Analysis Dashboard using Power BI**, is to analyze historical crime data and present it in a visually appealing and interactive manner. Power BI enables efficient data modeling, dynamic filtering, and real-time interaction, allowing users to explore crime patterns from multiple perspectives. The dashboard helps in identifying crime trends over time, high-crime locations, frequently occurring crime types, victim demographics, and case resolution status.

This project focuses on providing a comprehensive overview of crime statistics using charts, maps, KPIs, and slicers. By using interactive visuals, users can drill down into specific areas, time periods, or crime categories. The insights derived from this dashboard can assist authorities in resource allocation, strategic planning, crime prevention measures, and policy formulation. Overall, the project demonstrates how data visualization and analytics can enhance decision-making in the domain of public safety.

2. Source of Dataset

The dataset is from data.gov.in. The dataset used in this project is a publicly available **crime dataset** containing detailed records of reported crime incidents. The data includes attributes such as crime type, date and time of occurrence, location (latitude and longitude), victim details, weapon used, and case status. This dataset is suitable for exploratory data analysis and visualization-based insights.

3. Dataset Preprocessing

Dataset preprocessing is a critical step in any data analysis project, as the quality of insights largely depends on the quality of data used. The raw crime dataset contained inconsistencies, missing values, and redundant information, which required thorough cleaning and transformation before visualization in Power BI.

The following preprocessing steps were performed:

- **Removal of Duplicate Records:** Duplicate entries were identified and removed to avoid over-counting of crime incidents.
- **Handling Missing Values:** Records with missing or null values in important fields such as crime type, location, or date were examined. Wherever possible, missing values were handled appropriately to maintain data consistency.
- **Standardization of Column Names:** Column names were renamed and standardized to improve readability and ease of use during analysis.
- **Date and Time Formatting:** Date and time attributes were formatted correctly to enable time-based analysis such as year-wise, month-wise, and trend analysis.
- **Categorical Data Validation:** Categorical columns such as crime type, weapon used, gender, and case status were checked for spelling errors and inconsistent labels, and corrected where necessary.
- **Geographical Data Validation:** Latitude and longitude values were verified to ensure accuracy for map-based visualization and hotspot analysis.

These preprocessing steps ensured that the dataset was clean, reliable, and analysis-ready. Proper preprocessing enhanced the accuracy of visualizations and helped in generating meaningful and trustworthy insights from the crime data.

4. Analysis on Dataset

The analysis phase focuses on extracting meaningful insights from the crime dataset based on clearly defined objectives. Each objective is supported by descriptive analysis and appropriate visualizations created using Power BI.

Objective 1: Overall Crime Overview

i. General Description

This objective aims to understand the overall volume and distribution of crime incidents recorded in the dataset. It provides a high-level summary of crime occurrences and helps in assessing the severity of crime in the observed region.

ii. Specific Requirements

- Total number of crime incidents
- Overall crime trend across different years or months

iii. Analysis Results

The analysis reveals variations in crime frequency across different time periods. Certain years or months show higher crime intensity, indicating periods of increased criminal activity. This information is useful for identifying seasonal or annual crime patterns.

iv. Visualization

KPI cards are used to represent total crime count, while line or column charts illustrate trends over time.

Objective 2: Crime Type Analysis

i. General Description

This objective focuses on analyzing different categories of crimes reported in the dataset to identify the most frequent crime types.

ii. Specific Requirements

- Crime count for each crime category

iii. Analysis Results

The analysis shows that certain crime types occur more frequently than others, contributing significantly to the overall crime rate. These crime categories require greater attention from law enforcement agencies.

iv. Visualization

Bar charts and donut charts are used to represent crime-wise distribution clearly.

Objective 3: Location-wise Crime Distribution

i. General Description

This objective identifies geographical areas with high crime concentration.

ii. Specific Requirements

- Area-wise or location-wise crime count

iii. Analysis Results

The analysis highlights specific locations that consistently report higher crime incidents, indicating potential crime hotspots.

iv. Visualization

Map visuals using latitude and longitude effectively display crime density across regions.

Objective 4: Time-based Crime Pattern Analysis

i. General Description

This objective examines how crime varies over time.

ii. Specific Requirements

- Year-wise, month-wise, or time-wise crime trends

iii. Analysis Results

Crimes show noticeable variation during certain time periods, which may be influenced by social or environmental factors.

iv. Visualization

Line charts and column charts are used to show temporal patterns.

Objective 5: Victim and Gender Analysis

i. General Description

This objective focuses on analyzing victim demographics to understand crime impact.

ii. Specific Requirements

- Gender-wise distribution of victims

iii. Analysis Results

The analysis indicates differences in victimization across demographic groups, helping identify vulnerable populations.

iv. Visualization

Donut charts and bar charts represent demographic distribution.

Objective 6: Weapon Used Analysis

i. General Description

This objective analyzes the types of weapons involved in crimes.

ii. Specific Requirements

- Weapon-wise crime count

iii. Analysis Results

The results show that certain weapons are more frequently involved in crimes, indicating higher risk factors.

iv. Visualization

Bar charts are used to compare weapon usage.

Objective 7: Case Status Analysis

i. General Description

This objective evaluates the status of reported crime cases.

ii. Specific Requirements

- Solved, pending, and under-investigation cases

iii. Analysis Results

A significant number of cases remain unresolved, highlighting the need for improved investigation strategies.

iv. Visualization

Donut charts and KPI visuals show case resolution status.

5. Conclusion

The **Crime Lens dashboard** successfully transforms raw crime data into meaningful insights. By analyzing crime trends, locations, victim details, and case status, the project provides a comprehensive overview of crime patterns. The interactive nature of the

dashboard allows users to explore data dynamically, making it a powerful decision-support tool.

6. Future Scope

- Integration of real-time crime data
- Predictive crime analysis using machine learning
- Area-wise crime forecasting
- Integration with demographic and socio-economic data
- Enhanced drill-through and alert systems

7. References

- [1] Microsoft Power BI Documentation
- [2] Crime Data Analysis Research Papers
- [3] IEEE Data Visualization Standards